

Md Minhazul Islam

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🌐 Portfolio

🌐 LinkedIn

🐙 GitHub

🏆 Kaggle

Rajshahi, Bangladesh

About Me

Md Minhazul Islam is a final-year BSc student in Computer Science and Engineering at Rajshahi University of Engineering and Technology. Passionate about advancing computational techniques, his research spans Natural Language Processing, Medical Text Analysis, Computational Linguistics, and Machine Learning. With a strong foundation in AI-driven solutions, he has contributed to innovative projects and published extensively in conferences and journals. He is a hard worker and determined to take any challenge.

Research Interests

My research focuses on leveraging computational methods to address complex challenges in language and data analysis. Key areas include:

- Natural Language Processing (NLP) and Computational Linguistics
- Medical Text Analysis and Healthcare Informatics
- Multimodal Learning and Contextual Embeddings
- Explainable AI and Model Interpretability
- Topic Modeling and Semantic Search
- Bengali Linguistics and Low-Resource Language Processing

Publications

Below is a selection of my peer-reviewed publications, reflecting contributions to NLP, computational linguistics, and AI-driven analytics.

- [C1] **Islam, M. M.**, Jubaer, M. T., Srizon, A. Y. (2023). *Navigating Bengali Linguistics: Insights from Machine and Deep Learning Perspectives for Categorization of Sentences*. ICCIT 2023. [Code, Methodology]
- [C2] **Islam, M. M.**, Jubaer, M. T., Rashid, M. H. O. (2024). *Consistency of Contextual Embedding in Literary Texts*. ICISSET 2024. [Main Idea, Code, Methodology]
- [C3] Jubaer, M. T., Chowdhury, B., **Islam, M. M.**, Rashid, M. H. O. (2024). *Detecting Discrepancy to Enhance the Accuracy of Sentiment and Emotion Analysis*. STI 2024. [Coding, Abstract]
- [C4] Jubaer, M. T., Farzana, M., Chowdhury, B., Zaman, M. S. U., Srizon, A. Y., **Islam, M. M.** (2024). *Analyzing Audience Engagement in Esports: Sentiment and LLM-Based Topic Insights*. ICCIT 2024. [Literature, Revision]
- [C5] **Islam, M. M.**, Jubaer, M. T., Srizon, A. Y., Hossain, M. A. (2024). *Explainable AI-Driven Improved Disease Prediction through Symptom Analysis*. ICCIT 2024. [Idea, Code, Methodology, Conclusion]
- [J1] Rashid, M. H. O., Jubaer, M. T., **Islam, M. M.**, et al. (2024). *Live and Mediated User Engagements: A Comparative Dataset from Two Bengali Audio-Story-Based YouTube Channels*. Data in Brief, Elsevier. [Data Translation, Revision]

Experience

• Young Learners Research Lab

Research Assistant

Jan 2023 – Present

Remote

- Conducted research on NLP and deep learning, enhancing model efficiency and explainability.
- Focused on Bengali and English datasets, addressing underexplored linguistic domains.
- Co-authored multiple conference and journal publications, with ongoing research projects.

• Computational Linguistics Research Lab

Research Assistant

Jun 2023 – Present

Remote

- Investigated word embeddings, contextual embeddings, and topic modeling in literary texts.
- Analyzed linguistic patterns in American slave narratives and performed stylometric analysis.

– Developed novel NLP techniques for literary research and computational approaches.

● **Outlier**

AI Trainer (Mathematics and Coding, Contributor & Quality Control)

Feb 2024 – Present

Remote

– Contributed to dataset creation and refined AI model outputs in coding and mathematics.

– Performed rigorous quality control on AI-generated content for accuracy.

● **Private Tutor**

HSC and University Students

2020 – Present

Rajshahi, Bangladesh

– Tutored HSC students in Physics, Mathematics, and ICT, and university students in Object-Oriented Programming.

Education

● **Rajshahi University of Engineering and Technology**

BSc in Computer Science and Engineering

Feb 2020 – Jun 2025

Rajshahi, Bangladesh

– CGPA: 3.39/4.00 (up to 6th semester)

● **Rajshahi Govt. City College**

Higher Secondary Certificate

May 2017 – Apr 2019

Rajshahi, Bangladesh

– GPA: 5.00/5.00

● **Sankoa High School**

Secondary School Certificate

Jan 2015 – Mar 2017

Bagmara, Rajshahi, Bangladesh

– GPA: 5.00/5.00

Projects

● **Contextual Sarcasm Detection (Undergraduate Thesis)**

Python, PyTorch, Transformers, Pandas, NumPy

2025



– Developed an encoder-encoder cross-attention model for contextual sarcasm detection.

– Combined CLIP for image encoding and mBERT for text processing, achieving 95% accuracy.

– Currently refining a custom multimodal architecture for enhanced performance.

● **Clickbait Content Detection**

Python, Scikit-learn, PyTorch, Transformers, CLIP, Pandas, NumPy

2023–2024



– Built a tool to detect clickbait on social media platforms using multimodal analysis.

– Integrated CLIP and mBERT, achieving 95% accuracy in initial evaluations.

– Ongoing data collection and model refinement for improved reliability.

● **Medical Triage Classification**

Python, Transformers, Scikit-learn, Pandas, NumPy, Matplotlib

2024–2025



– Developed a custom BERT-based model for classifying medical triage notes.

– Designed regex and medical dictionary for text preprocessing, achieving 83% accuracy.

– Explored multimodal and EDBERT models for enhanced classification performance.

● **US Election Data Analysis**

Python, PyTorch, LLM, PCA, Topic Modeling, Transfer Learning

2025



– Analyzed Twitter posts using topic modeling, BERT-based semantic search, and GraphRAG.

– Clustered embeddings with t-SNE and HDBSCAN to extract election-related insights.

– Developed an alternative to traditional exit polls, with ongoing evaluation.

Skills

Programming	Python, C, C++, PHP
Data Science & ML	PyTorch, Scikit-learn, TensorFlow, Keras, Pandas, NumPy, Matplotlib, Seaborn, XGBoost
Transformers & LLMs	BERT, T5, LLM, Fine-tuning, PEFT, Quantization, LoRA, RAG, Prompt Engineering
Cloud & DevOps	AWS, Docker, GitHub
Specialized Areas	NLP, Text Analysis, Semantic Search, Topic Modeling, Contextual Embeddings, Deep Learning
Tools	Jupyter Notebooks, OpenCV, Hugging Face, SQL
Research	Literature Review, Data Collection, Model Development, Statistical Analysis

Honors and Awards

- **Top 15 in Dataverse Challenge** 2023
IT Verse 
 - Ranked in the top 15, demonstrating strong data analysis and ML skills.
- **Top 10 in CUET ETE DAY 2023 ML Competition** 2023
CUET 
 - Secured top 10 position, showcasing expertise in model building and optimization.
- **Finalist in Robi Datathon 3.0** May 2024
Robi Headquarters, Dhaka 
 - Reached the final round, showcasing data science and analytics skills.

Leadership Experience

- **Officer Cadet** Jan 2022 – Apr 2022
Bangladesh Military Academy 
 - Completed a 3-month training program in military discipline, tactics, and leadership.
 - Developed skills in teamwork, resilience, and time management under pressure.
- **Team Leader - Robi Datathon 3.0 Finalist** May 2024
Robi Headquarters, Dhaka 
 - Led a team to the final round, developing a data-driven business solution.
 - Coordinated efforts to ensure efficient problem-solving and innovation.

Volunteer Experience

- **Social Worker** 2024 – Present
BD Clean Bagmara, Rajshahi 
 - Raised awareness on environmental conservation, cleanliness, and waste management.
 - Participated in community events to promote health and hygiene practices.
- **AI Research Mentor** 2023 – Present
University Research Community 
 - Mentored students in AI, ML, and data analysis, organizing knowledge-sharing sessions.

Certifications

- **Green Card – Eligible for Military Officer Course** Dec 2021
Inter Services Selection Board, Bangladesh
- **Certificate of Appreciation - 26th Int. Conf. On Computer and Information Technology** 2023 *Organizer: IEEE Bangladesh*

- **Conference Paper Presentation (ICISSET)**

International Islamic University Chittagong

2024

- **Conference Paper Presentation (ICCIT-2024)**

IEEE Bangladesh Section

Dec 2024

Additional Information

Languages: English (Intermediate), Hindi (Intermediate)

Interests: Football, Traveling, Fitness

References

1. **Mohammad Harun Or Rashid, PhD**

Assistant Professor, Department of English, RUET

Email: harunmr9@gmail.com

Relationship: Research Supervisor (Text Analytics, Digital Humanities)

2. **Md. Azmain Yakin Srizon**

Assistant Professor, Department of CSE, RUET

Email: azmainsrizon@cse.ruet.ac.bd

Relationship: Undergraduate Thesis Supervisor

3. **Dr. Md. Ali Hossain**

Professor, Department of CSE, RUET

Email: aliruet@gmail.com

Relationship: Research Supervisor (Medical Text Analysis)